Linear Bandits

Monday, June 2, 2025 9:17 AM

n arms i. Arm i yields a reward

$$\theta^{*T} \phi_i + w$$

where ϕ_i is a known feature vector, θ is a weight vector that has to learned, and w is random noise assumed to be 1-subG. The noise is assumed to be independent for each pull.

$\theta^*, \phi_i \in \mathbb{R}^d$. We are interested in the case $d << n$ and we hope to get regret bounds that depend on d and not n.

Let A_j denote the jth arm that is pulled and r_j be the corresponding reward. Then after t pulls, suppose we estimate θ using regularized least squares:

$$\min_{\theta} \frac{1}{2} \sum_{j=1}^{t} \| \theta^T \phi_{A_j} - R_j \|^2 + \frac{\lambda}{2} \|\theta\|^2$$

$$\Rightarrow \sum_{j=1}^{t} (\theta^T \phi_{A_j} - R_j) \phi_{A_j} + \lambda \theta = 0$$

$$\left(\lambda I + \sum_{j=1}^{t} \phi_{A_j} \phi_{A_j}^T\right) \theta = \sum_{j=1}^{t} r_j \phi_{A_j}$$

Or

$$\hat{\theta}_t = \underbrace{\left(\lambda I + \sum_{j=1}^{t} \phi_{A_j} \phi_{A_j}^T\right)^{-1}}_{\triangleq V_t^{-1}} \left(\sum_{j=1}^{t} r_j \phi_{A_j}\right)$$

Note regularizing with $\frac{\lambda \|\theta\|^2}{2}$ makes sure this term is invertible

$$\Rightarrow \quad \hat{\theta}_t = V_t^{-1} \left(\sum_{j=1}^{t} r_j \phi_{A_j}\right)$$

Define C_t to be following set:

$$C_t = \left\{ \theta : \|\theta - \hat{\theta}_{t-1}\|^2_{V_{t-1}} \le \beta_t \right\},$$

where $\beta_t \ge 1$ will be chosen later

So our UCB algorithm for linear bandits is:
At time t, pull the arm a with the largest

$$\max_{\theta \in C_{t-1}} \theta^T \phi_a$$

For now, we will assume β_t has been chosen s.t.

For now, we will assume β_t has been chosen s.t. $\theta^* \in \mathcal{C}_t$ w.p. $\geq 1-\delta$. Further, we will assume

(a) $|\theta^{*T}\phi_a| \leq 1 \quad \forall a$

(b) $\|\phi_a\|_2 \leq L \quad \forall a$ and $\|\theta^*\| \leq L$

Regret analysis: (w.l.o.g. assume 1 is the best arm)

Let $r_t = \theta^{*T}(\phi_1 - \phi_{A_t})$ be the instantaneous regret.

w.p. $\geq 1-\delta$, $\theta^* \in \mathcal{C}_{t-1}$, hence

$$\theta^{*T}\phi_1 \leq \max_{\theta \in \mathcal{C}_t} \theta^T \phi_1 \leq \max_{\theta \in \mathcal{C}_t} \theta^T \phi_{A_t}$$

A_t is chosen s.t. this is true (call the maximizer $\tilde{\theta}$)

$$\theta^{*T}\phi_1 - \theta^{*T}\phi_{A_t} \leq \tilde{\theta}\phi_{A_t} - \theta^{*T}\phi_{A_t}$$

$$\leq \|\tilde{\theta} - \theta^*\|_{V_{t-1}} \|\phi_{A_t}\|_{V_{t-1}^{-1}}$$

C-S ↑

$$\leq \sqrt{\beta_t}\, \|\phi_{A_t}\|_{V_{t-1}^{-1}}$$

$$\Rightarrow \quad r_t = \min\left(2, \sqrt{\beta_t}\, \|\phi_{A_t}\|_{V_{t-1}^{-1}}\right)$$

↑ by assumption $|\theta^{*T}\phi_1| \leq 1$ + $|\theta^{*T}\phi_a| \leq 1 \ \forall a$

$$\Rightarrow \quad r_t = \sqrt{\beta_t}\,\min\left(\frac{2}{\sqrt{\beta_t}}, \|\phi_{A_t}\|_{V_{t-1}^{-1}}\right)$$

$$\leq \sqrt{\beta_t}\,\min\left(2, \|\phi_{A_t}\|_{V_{t-1}^{-1}}\right)$$

$$\leq \sqrt{\beta_t}\,\min\left(2, 2\|\phi_{A_t}\|_{V_{t-1}^{-1}}\right)$$

$$= 2\sqrt{\beta_t}\,\min\left(1, \|\phi_{A_t}\|_{V_{t-1}^{-1}}\right)$$

$$\Rightarrow \text{ w.p. } \geq 1-\delta$$

$$R_T = \sum_{t=1}^{T} r_t \leq \sqrt{T \sum_{t=1}^{T} r_t^2}$$

$$\leq 2\sqrt{T \sum_{t=1}^{T} \min\left(1, \|\phi_{A_t}\|^2_{V_{t-1}^{-1}}\right)\beta_t}$$

- This is the key result. The rest of the proof is about bounding the term

$$\sum_{t=1}^{T} \min\left(1, \|\phi_{A_t}\|^2_{V_{t-1}^{-1}}\right)\beta_t$$

as a function of T.

- We first work on β_t

From the lecture on concentration for least squares chose

$$\sqrt{\beta_t} = \sqrt{\lambda}\|\theta^*\| + \sqrt{\log\frac{\det(V_{t-1})}{\lambda^d} + 2\log\left(\frac{1}{\delta}\right)},$$

$$\sqrt{\beta_t} = \sqrt{\lambda}\|\theta^*\| + \sqrt{\log \frac{\det(V_{t-1})}{\lambda^d} + 2\log\left(\frac{1}{\delta}\right)},$$

then $\theta^* \in \mathcal{C}_{t-1}$ w.p. $\geq 1-\delta$.

$$\det(V_{t-1})^{1/2} = \left(\prod_{i=1}^{d} \lambda_i(V_{t-1})\right)^{1/d} \overset{\text{AM-GM}}{\leq} \frac{1}{d}\sum_{i=1}^{d} \lambda_i(V_{t-1})$$

$$= \frac{1}{d}\operatorname{Tr}(V_{t-1})$$

$$\operatorname{Tr}(V_{t-1}) = \operatorname{Tr}\left(\lambda I + \sum_{s=1}^{t-1} \phi_{A_s}\phi_{A_s}^T\right)$$

$$= \operatorname{Tr}(\lambda I) + \sum_{s=1}^{t-1}\|\phi_{A_s}\|^2$$

$$\leq d\lambda + (t-1)L^2$$

$$\det(V_{t-1}) \leq \left(\lambda + \frac{(t-1)L^2}{d}\right)^d$$

So, β_t can be chosen more conservatively as

$$\sqrt{\beta_t} \leq \sqrt{\lambda}L + \sqrt{\log\frac{\left(\lambda + \frac{(t-1)L^2}{d}\right)^d}{\lambda^d} + 2\log\left(\frac{1}{\delta}\right)}$$

$$\leq \underbrace{\sqrt{\lambda}L + \sqrt{d\log\frac{\left(\lambda + \frac{(T-1)L^2}{d}\right)}{\lambda} + 2\log\left(\frac{1}{\delta}\right)}}_{C_T}$$

$$\Rightarrow \quad R_T \le 2\sqrt{T C_T^2 \sum_{t=0}^{T-1} \min\left(1, \|\phi_{A_t}\|^2_{V_{t-1}^{-1}}\right)}$$

Next, we work on the sum in the above expression.

Further, since

$$\|\theta^*\| \le L. \quad \text{and}$$

$$V_{t-1} = \lambda I + \sum_{s=1}^{t-1} \phi_{A_s} \phi_{A_s}^T$$

$$\Rightarrow \quad V_t = V_{t-1} + \phi_{A_t} \phi_{A_t}^T$$

$$= V_{t-1}^{1/2} V_{t-1}^{1/2} + \phi_{A_t} \phi_{A_t}^T$$

$$= V_{t-1}^{1/2} \left(V_{t-1}^{1/2} + V_{t-1}^{-1/2} \phi_{A_t} \phi_{A_t}^T \right)$$

$$= V_{t-1}^{1/2} \left(I + V_{t-1}^{-1/2} \phi_t \phi_t^T V_{t-1}^{-1/2} \right) V_{t-1}^{1/2}$$

$$\det(V_t) = \det\left(V_{t-1}^{1/2}\right) \det\left(I + V_{t-1}^{-1/2} \phi_t \phi_t^T V_{t-1}^{-1/2}\right) \det\left(V_{t-1}^{1/2}\right)$$

$$= \det(V_{t-1}) \det\left(I + \underbrace{V_{t-1}^{-1/2} \phi_t \phi_t^T V_{t-1}^{-1/2}}\right)$$

eigenvalues are 0 and $\|\phi_t\|^2_{V_{t-1}^{-1}}$

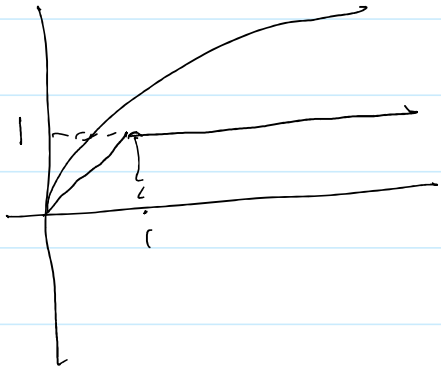
$$= \det(V_{t-1}) \left(1 + \|\phi_t\|^2_{V_{t-1}^{-1}}\right)$$

$$\Rightarrow \det(V_t) = \det(\underbrace{V_0}_{\lambda I}) \prod_{s=1}^{t} \left(1 + \|\phi_s\|^2_{V_{s-1}^{-1}}\right)$$

$$\Rightarrow \sum_{s=1}^{t} \log\left(1 + \|\phi_s\|^2_{V_{s-1}^{-1}}\right) = \log \frac{\det(V_t)}{\lambda^d}$$

Further,

$$\min(1, y) \leq 2\log(1+y)$$



$$\Rightarrow \sum_{s=1}^{t-1} \min\left(1, \|\phi_{A_t}\|_{V_{s-1}^{-1}}\right) \leq \log \frac{\det(V_t)}{\lambda^d}$$

$$\leq d \log\left(1 + \frac{tL^2}{\lambda d}\right)$$

This is called the elliptic potential lemma

Putting it all together,

$$R_T \leq 2\sqrt{T C_T^2 \, 2 \log \frac{\det(V_T)}{\lambda^d}}$$

$$= \sqrt{8 T G_T^2 d \log\left(1 + \frac{T L^2}{\lambda d}\right)}$$

since $G_T^2 = d$

$$R_T = O\left(d\sqrt{T}\right) \quad \text{w.p.} \geq 1-\delta$$

$$\overline{R_T} = O\left(d\sqrt{T}\right)(1-\delta) + \delta \Delta T$$

(Δ: worst-case gap between best and another arm)

$$= \tilde{O}\left(d\sqrt{T}\right) \quad \text{by choosing } \delta = \frac{1}{\sqrt{T}}$$

↓

ignoring $\log T$ terms.